

# Amrit Raj

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## Experience

### Pinnacle Labs

Remote

Web Development Intern

June – July 2026

- Engineered a full-stack **event management platform** on **Next.js** and **TypeScript**, enabling organizers to create, publish, and manage events end-to-end
- Resolved booking race conditions using **row-level locking**, and implemented **idempotent APIs** to prevent redundant calls, ensuring reliable event registrations under concurrent requests
- Designed and implemented **RESTful APIs** with **Drizzle ORM**, enforcing atomic transactions for reliable data integrity
- Automated **email ticket and reminder delivery** on event creation, streamlining outreach and eliminating manual invite tracking

## Technical Skills

- Languages:** JavaScript, TypeScript, Java
- Backend:** Node.js, Express.js, REST APIs, JWT, WebSockets
- AI/ML:** LangChain, RAG, Vector Databases, OpenRouter, Inngest
- Databases & ORM:** PostgreSQL, Drizzle, Redis, Pinecone
- Frontend:** React.js, Next.js, HTML, CSS
- Infrastructure:** Docker
- Developer Tools:** Git, GitHub, Postman
- Core Subjects:** DSA, OS, DBMS, OOP, Computer Networks

## Projects

### PRRabbit – AI-Powered GitHub PR Reviewer

[Live](#) [GitHub](#)

- Architected an **automated PR review system** as a GitHub App, using **Inngest** for async job orchestration to decouple webhook response times from long-running AI inference
- Implemented **diff parsing with line-level mapping**, translating raw unified diffs into addressable code positions for accurate inline commenting via the **GitHub Reviews API**
- Integrated **Pinecone-backed RAG retrieval** to pull relevant repository context into each review prompt, grounding feedback in the actual codebase instead of the diff alone
- Abstracted model access through **OpenRouter**, decoupling the review pipeline from any single LLM provider

### AuthCraft – OIDC Identity Provider

[Live](#) [GitHub](#)

- Engineered a **spec-compliant OpenID Connect Identity Provider** from scratch, implementing the full **Authorization Code Flow** with **RS256 JWT signing** and asymmetric RSA key pair management
- Implemented **PKCE with S256 verification** and refresh token rotation with reuse detection, eliminating token replay attacks across all client sessions
- Exposed standard **JWKS** and **OIDC Discovery endpoints**, enabling autonomous token verification by any OIDC-compliant client
- Containerized full stack with **Docker Compose** (Node.js + PostgreSQL), reducing environment setup from 30 minutes to under 2 minutes

### Muzzix – Collaborative Music Queue App

[Live](#) [GitHub](#)

- Built a real-time collaborative music queue app where users join shared rooms, add songs, upvote songs, and vote to control playback order collectively – synced live across all connected clients
- Eliminated audio drift via **NTP-style clock sync**, keeping playback skew under 500ms across all connected clients
- Reduced vote queue processing from **O(N log N)** to **O(log N)** by switching to Redis Sorted Sets, removing per-vote server recalculation
- Achieved **sub-20ms event propagation** across scaled WebSocket servers using Redis Pub/Sub broadcast architecture
- Resolved concurrent vote race conditions with **Redis atomic operations**, guaranteeing correct counts under simultaneous writes with no locking overhead

## Achievements

- Solved 800+ DSA problems** across LeetCode and GeeksforGeeks, maintaining consistent active coding practice
- Currently **serving as Webmaster** for the IEEE Student Branch at AIMT, managing and maintaining the official website

## Education

### Ambalika Institute of Management and Technology

B.Tech in Computer Science and Engineering

Lucknow, India

2023 – 2027